Functional Annotation Report: LigD (Q88HU3, PP_3260) — *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

The gene symbol **ligD** is **unambiguous** and correctly matches the UniProt description for the target.

- **UniProt:** Q88HU3 — "DNA ligase (ATP)" (EC 6.5.1.1); AltName "NHEJ DNA polymerase"
- **Gene:** *ligD*; ordered locus **PP_3260**; organism *Pseudomonas putida* KT2440 (taxid 160488)
- **Length:** 833 aa (verified from UniProt sequence, Iteration 2)
- **Domains (InterPro):** LigD_PE_domain (IPR014144), LigD_ligase_dom (IPR014146), DNA_ligase_ATP-dep central/C (IPR012310/IPR012309), LigD_pol_dom (IPR014145)

All identifiers, the enzyme class, the domain complement, and the organism are internally consistent and match the well-characterized bacterial NHEJ **DNA ligase D** family. A study performed directly in *P. putida* (Paris et al., 2015, P 25942369) explicitly describes the LigD/Ku NHEJ system in this organism, and the cognate Ku gene is present in the same genome (PP_3255; see below). There is **no evidence of gene-symbol ambiguity** — this is the canonical bacterial NHEJ multifunctional ligase, not an unrelated same-symbol gene.

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1. Summary (Answer to the Research Question)

LigD is the central, multifunctional, ATP-dependent DNA ligase of the bacterial non-homologous end-joining (NHEJ) pathway. Its primary function is to repair chromosomal DNA **double-strand breaks (DSBs)** without a homologous template, acting as the enzymatic "workhorse" downstream of the DNA-end-binding protein **Ku**. LigD is a single polypeptide carrying **three autonomous catalytic modules** that together perform every chemical step of end-joining: a **3'-phosphoesterase (PE)** domain that heals damaged 3' ends to a ligatable 3'-OH; an **ATP-dependent ligase (LIG)** domain that seals the phosphodiester backbone via a covalent ligase–AMP intermediate; and a **primase/polymerase (POL)** domain that fills gaps and adds (ribo)nucleotides across the break. The enzyme functions in the **cytoplasm on nucleoid (chromosomal) DNA**, and is physiologically most important in **non-replicating / stationary-phase / starved cells**, where homologous recombination is unavailable. NHEJ by Ku–LigD is intrinsically **error-prone**, so LigD also contributes to stress-associated (stationary-phase) mutagenesis in *P. putida*.

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2. Primary Function: Reaction Catalyzed and Substrate Specificity

2.1 Overall pathway function

Bacterial NHEJ is a minimal **two-component system**: the homodimeric end-binding protein **Ku** plus the polyfunctional ligase **LigD** possess all the break-recognition, end-processing and ligation activities needed to rejoin a DSB (Gong et al., 2005, P 15778718; Pitcher, Brissett & Doherty, 2007, P 17506672; Amare et al., 2021, P 34901162). Ku binds and synapses the two broken ends and recruits/stimulates LigD, which then processes and seals the junction (Amare et al., 2021, P 34901162; Zhu & Shuman, 2010, P 20018881).

2.2 Domain architecture (bioinformatically verified for Q88HU3)

InterPro/Pfam mapping of the 833-aa Q88HU3 sequence gives an unambiguous N→C module order, matching the experimentally characterized *Pseudomonas aeruginosa* LigD:

Module	Approx. residues	InterPro / Pfam	Catalytic role
PE (3'-phosphoesterase)	~5–160	IPR014144 / PF13298	End-healing: generate 3'-OH
LIG (ATP-dependent ligase)	~219–520 (central IPR012310/ PF01068 + OB-fold C-term IPR012309/PF04679)	IPR014146	Nick sealing
POL (primase-polymerase)	~547–812 (PaeLigD-type IPR033651)	IPR014145 / PF21686	Gap fill / nucleotide addition

The POL module is explicitly of the **PaeLigD-type** subclass, i.e., the same structural class as the crystallized *P. aeruginosa* POL domain, allowing high-confidence transfer of the *P. aeruginosa* biochemistry to the *P. putida* ortholog (Iteration 2 analysis).

2.3 POL domain — gap-filling primase/polymerase

- **Fold/mechanism:** The 1.5-Å crystal structure of the *Pseudomonas* LigD POL domain reveals a **minimized two-metal polymerase with a fold similar to archaeal DNA primase** (archaeo-eukaryotic primase, AEP superfamily) (Zhu et al., 2006, P 16446439).
- **Activity/specificity:** Performs **templated and nontemplated primer extension**, is **Mn^{2+}-dependent**, and shows a **preference for adding ribonucleotides to blunt DNA ends** (Zhu et al., 2006, P 16446439 ; Pitcher et al., 2006, P 17174332).
- **Substrate recognition:** Efficient **gap filling** requires a **5'-phosphate** on the distal strand of the gap, conferring apparent processivity; residues His-553, Arg-556, Lys-566 mediate 5'-PO_4

recognition and Phe-603 stacks on the templating base (Zhu & Shuman, 2010, 20018881). A 5'-phosphate at the single-/double-strand junction is a principal signal for specific POL binding to NHEJ intermediates (Pitcher et al., 2006, 17174332).

- **Physiological consequence:** POL is the **direct catalyst of mutagenic (nontemplated single-nucleotide) additions** during NHEJ in vivo (Zhu et al., 2006, 16446439).

2.4 PE domain — 3'-end healing

The N-terminal PE module catalyzes **Mn-dependent 3'-phosphodiesterase and 3'-phosphomonoesterase** reactions at the primer-strand 3' end (Zhu & Shuman, 2006, 16540477). It acts as a **3'-exoribonuclease** that resects a 3'-terminal diribonucleotide to a ribonucleoside-3'-PO_4 (strictly requiring the 2'-OH of the penultimate ribose), and its phosphomonoesterase then **converts a 3'-PO_4 (ribo- or deoxyribo-) to a ligatable 3'-OH** (Zhu & Shuman, 2006, 16540477). This "heals" damaged ends and removes the ribonucleotides deposited by POL, coordinating the processing→sealing hand-off.

2.5 LIG domain — nick sealing (EC 6.5.1.1)

The ligase module is an **ATP-dependent DNA ligase** that performs the final sealing step. The structure of the LigD ligase domain was captured as the **covalent ligase–AMP (adenylylated-enzyme) intermediate** with a divalent metal in the active site (Akey et al., 2006, 16476729), consistent with canonical two-step ligase chemistry: (i) enzyme adenylylation using ATP, (ii) AMP transfer to the DNA 5'-phosphate, (iii) phosphodiester bond formation joining 3'-OH and

$5'\text{-}PO_4$. Its activity is dynamically balanced against end-remodeling, which underlies NHEJ's error-prone signature (Akey et al., 2006, 16476729).

Net reaction (EC 6.5.1.1): ATP + (5'-phospho-DNA)$_n$ + (3'-hydroxy-DNA)$_m$ → AMP + diphosphate + ligated DNA — carried out at DSB junctions after PE/POL end-processing.

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3. Subcellular Localization

LigD is a **soluble cytoplasmic enzyme** that acts on **chromosomal (nucleoid) DNA**. There is no signal peptide, transmembrane region, or secretion signal; the protein's substrate is intracellular double-stranded DNA. Functionally it is recruited to double-strand breaks in the bacterial chromosome, where Ku first binds the DNA ends and delivers LigD (Amare et al., 2021, 34901162; Zhu & Shuman, 2010, 20018881). Genetic work in *P. putida* demonstrates its action on the resident chromosome under starvation (Paris et al., 2015, 25942369).

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4. Biological Process and Physiological Context

- **When it matters:** NHEJ is "a critical repair mechanism when DNA is not replicating" (Amare et al., 2021, 34901162). It is especially important during **quiescent/stationary states** (late stationary phase, starvation, sporulation in spore-formers) when no sister chromosome is available for homologous recombination (Paris et al., 2015,

P 25942369

).

- **In *P. putida* specifically:** Genetic evidence shows **LigD and Ku participate in stationary-phase mutagenesis** in carbon-starved *P. putida*; deleting *ligD* or *ku* yields distinct mutation spectra, and **both the PE and POL domains** of LigD contribute to the mutational outcomes (Paris et al., 2015,

  P 25942369

).
- **Fidelity:** NHEJ is intrinsically **mutagenic** — repair of blunt and 5'-overhang DSBs proceeds with ~50% error rate, driven by templated fill-in, nontemplated single-nucleotide additions, and nucleolytic resection (Gong et al., 2005,

  P 15778718

).
  Genetic dissection shows POL activity mainly promotes infidelity, while efficient repair depends on the Ku protein and the LigD POL **domain** (as a structural bridging element) more than on its polymerase catalytic activity per se (Aniukwu, Glickman & Shuman, 2008,

  P 18281464

).
- **Genomic context:** The cognate **Ku** (Q88HU8, gene *ku*, **PP_3255**, 273 aa) is encoded within five ORFs of *ligD* (PP_3260), confirming a complete, self-standing two-component NHEJ apparatus in KT2440 (Iteration 2 UniProt analysis).

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5. Evidence Basis

- **Direct experimental (in-organism):** Genetic knockouts and mutation-spectrum analysis in *P. putida* (Paris et al., 2015,

  P 25942369

).
- **Biochemical/structural (Pseudomonas orthologs, directly transferable):** Crystal structure and enzymology of the *Pseudomonas* POL domain (Zhu et al., 2006,

  P 16446439

  ;
  Zhu & Shuman, 2010,

  P 20018881

  );

PE-domain enzymology (Zhu & Shuman, 2006, P 16540477).

- **Structural (family):** LigD ligase-domain structure (Akey et al., 2006, P 16476729); *M. tuberculosis* PolDom structure (Pitcher et al., 2006, P 17174332); in-vivo domain contributions (Aniukwu et al., 2008, P 18281464); founding genetics of bacterial NHEJ (Gong et al., 2005, P 15778718).
- **Reviews:** Pitcher, Brissett & Doherty, 2007 (P 17506672); Amare et al., 2021 (P 34901162).
- **Bioinformatic/sequence:** UniProt Q88HU3 (833 aa) and InterPro/Pfam domain mapping confirming the PE–LIG–POL, PaeLigD-type architecture (this study, Iteration 2).

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6. Supported and Refuted Hypotheses

Supported: 1. LigD is the multifunctional ATP-dependent DNA ligase of bacterial NHEJ (EC 6.5.1.1). ✔ 2. It contains three separable catalytic modules (PE, LIG, POL) in N→C order PE–LIG–POL. ✔ (bioinformatically confirmed for Q88HU3) 3. POL is an AEP/primase-fold, Mn-dependent, ribonucleotide-preferring gap-filling polymerase that senses 5'-phosphate. ✔ 4. PE is a 3'-phosphoesterase/exoribonuclease generating 3'-OH ends. ✔ 5. It acts in the cytoplasm on chromosomal DNA, primarily in non-replicating/starved cells, and contributes to error-prone stationary-phase mutagenesis in *P. putida*. ✔ 6. A cognate Ku partner is genomically encoded (PP_3255) — complete two-component system. ✔

Refuted / not supported: - The gene-symbol-ambiguity concern is **refuted** — *ligD* unambiguously denotes the NHEJ ligase for this accession/organism.

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7. Limitations and Future Directions

- Most **atomic/enzymatic** data derive from *P. aeruginosa* and *M. tuberculosis* LigD; while domain identity (PaeLigD-type) makes transfer to *P. putida* robust, KT2440-specific in-vitro kinetics and a full-length structure are not yet reported.
- No experimental wild-type full-length LigD structure exists for any species; domain arrangement in the intact protein relies on individual-domain structures and AlphaFold predictions (Amare et al., 2021, P 34901162).
- The precise contribution of LigD to survival after specific DSB-inducing stresses (e.g., desiccation, ionizing radiation, quinolone antibiotics) in *P. putida* has not been exhaustively quantified and is a promising direction.
- Whether LigD/Ku influence horizontal gene transfer or plasmid capture via end-joining in *P. putida* remains open.

Report generated across Iterations 1–3. Citations reference PubMed IDs of the primary literature and reviews surveyed.
